

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and maintained detailed **General Arrangement drawings**, plans, and sections using **AutoCAD** for public realm schemes and transport infrastructure projects.
- Produced **3D models** for development masterplans through proficient use of **AutoCAD Civil 3D** and **Revit**, adhering to BIM standards.
- Collaborated with senior technicians and engineers to understand project requirements and contribute to design option discussions for transportation planning challenges.
- Assisted in the technical development of other team members by providing guidance on drawing production and software capabilities within the transportation team.
- Contributed to the refinement of good practice and processes by actively participating in the **Arup skills network** and sharing insights on technical design.
- Containerized multiple Java Spring Boot microservices with **Docker** and orchestrated deployments on an **AWS EKS** cluster, improving service resilience and manageability.
- Automated routine data extraction and reporting tasks using **Python scripting**, reducing manual effort for the operations team by approximately 8 hours per week.

SKILLS

CAD & BIM Software – AutoCAD, AutoCAD Civil 3D, Revit

Programming Languages – Java, Python, SQL

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Development Tools – Docker, Git, Jenkins

Databases – PostgreSQL, MySQL, Oracle

Methodologies – Agile, Scrum

PROJECTS

Transportation Design Software Integration

- Engineered Python scripts to automate data import and export between **AutoCAD Civil 3D** and a custom reporting tool, streamlining the generation of project summaries.
- Developed a series of **Java Spring Boot** microservices to process geometric data from transport models, providing validation and transformation capabilities for downstream analysis.
- Implemented unit tests using **JUnit** for the Java services, achieving over 85% code coverage and ensuring the reliability of data processing logic.

Automated Reporting and Analysis Pipeline

- Built a **Python**-based data pipeline to ingest and process transportation survey data, performing cleaning and aggregation using **pandas** and **NumPy**.
- Created interactive dashboards using **Power BI** to visualize key transportation metrics and trends, enabling stakeholders to quickly interpret project impacts.
- Wrote complex **SQL** queries against a PostgreSQL database to extract and validate data for the reporting pipeline, ensuring data integrity.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)